Start the class Docker container

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Prerequisites

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Before you begin, make sure you have:

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Before you begin, make sure you have:

1. **Docker Desktop** installed on your computer.

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1. **Docker Desktop** installed on your computer.
    - Download it from Docker Desktop.

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   - Follow the installation instructions for your operating system.

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1. **Docker Desktop** installed on your computer.
   - Download it from Docker Desktop.
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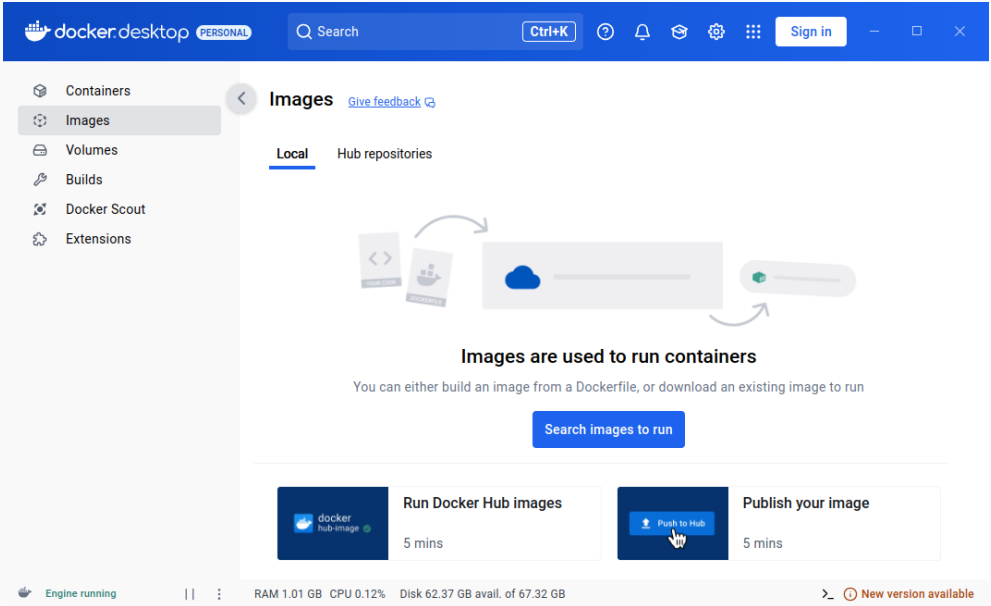
Open Docker Desktop and navigate to the “Images” tab:


docker desktop PERSONAL
Search
Ctrl+K
Sign in
Containers
Images
Volumes
Builds
Docker Scout
Extensions
Images
Give feedback
Local
Hub repositories
Images are used to run containers
You can either build an image from a Dockerfile, or download an existing image to run
Search images to run
Run Docker Hub images
5 mins
Publish your image
5 mins
Engine running
RAM 1.01 GB CPU 0.12% Disk 62.37 GB avail. of 67.32 GB
New version available

Prerequisites

2. **The Dockerfile** provided for the class.

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Save this in a file named `Dockerfile`:

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Save this in a file named `Dockerfile`:

```
FROM ubuntu:24.04
ENV DEBIAN_FRONTEND=noninteractive
RUN apt-get update && apt-get install -y --no-install-recommends \
    python3 python3-pip \
    bash openssh-server
RUN mkdir /var/run/sshd && \
    echo 'root:rootpassword' | chpasswd
EXPOSE 22
CMD ["/usr/sbin/sshd", "-D"]
```

Step 1: Build the Docker image

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   - Navigate to the directory where your `Dockerfile` is saved.

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   - Use the terminal to build the image:

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```
docker build -t class-docker .
```

Step 1: Build the Docker image

1. **Open a terminal**:
   - Navigate to the directory where your `Dockerfile` is saved.
2. **Run the build command**:
   - Use the terminal to build the image:

```
docker build -t class-docker .
```

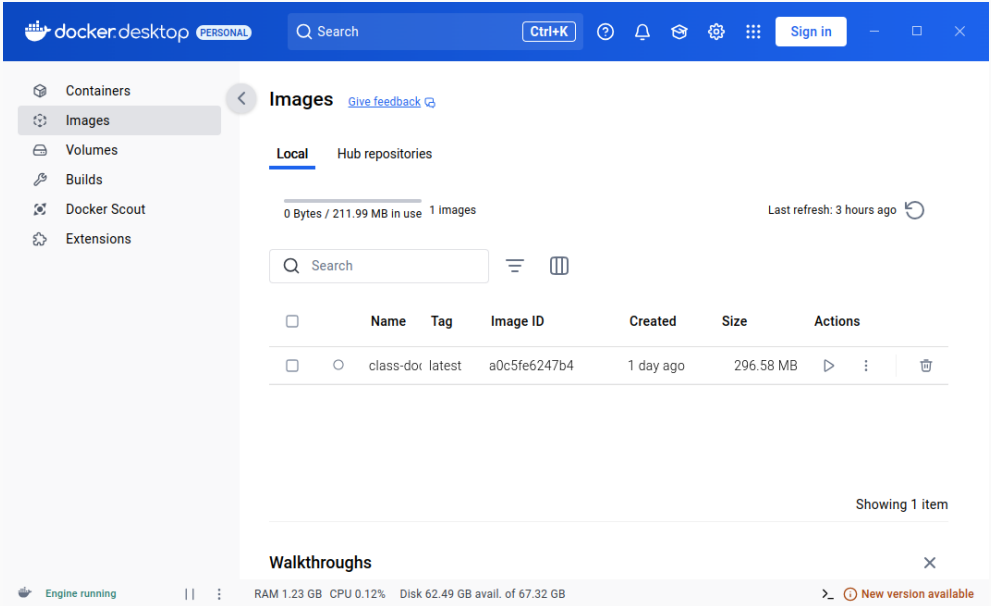
3. **Verify the image**:

Step 1: Build the Docker image

1. **Open a terminal**:
   - Navigate to the directory where your `Dockerfile` is saved.
2. **Run the build command**:
   - Use the terminal to build the image:

```
docker build -t class-docker .
```

3. **Verify the image**:
   - In the "Images" tab, check that `class-docker` is listed.


docker desktop PERSONAL
Search
Ctrl+K
Sign in
Containers
Images
Volumes
Builds
Docker Scout
Extensions
Images
Give feedback
Local
Hub repositories
0 Bytes / 211.99 MB in use
1 Images
Last refresh: 3 hours ago
Search
Name
Tag
Image ID
Created
Size
Actions
class-doc
latest
a0c5fe6247b4
1 day ago
296.58 MB
Showing 1 item
Walkthroughs
Engine running
RAM 1.23 GB CPU 0.12%
Disk 62.49 GB avail. of 67.32 GB
New version available

Step 2: Run the Docker Container

1. **Launch the container** from Docker Desktop:

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1. **Launch the container** from Docker Desktop:
   - Go to the “Images” tab.

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1. **Launch the container** from Docker Desktop:
   - Go to the "Images" tab.
   - Click the play icon next to the `class-docker` image.

Step 2: Run the Docker Container

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   - Click the play icon next to the `class-docker` image.
   - Set the following options:

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1. **Launch the container** from Docker Desktop:
   - Go to the "Images" tab.
   - Click the play icon next to the `class-docker` image.
   - Set the following options:
     - **Name**: `class-container`

Step 2: Run the Docker Container

1. **Launch the container** from Docker Desktop:
   - Go to the "Images" tab.
   - Click the play icon next to the `class-docker` image.
   - Set the following options:
     - **Name**: `class-container`
     - **Ports**: Map "Container port" 22 to "Host port" 2222.

Step 2: Run the Docker Container

1. **Launch the container** from Docker Desktop:
   - Go to the "Images" tab.
   - Click the play icon next to the `class-docker` image.
   - Set the following options:
     - **Name**: `class-container`
     - **Ports**: Map "Container port" 22 to "Host port" 2222.
     - **Volumes**: In "Host path," enter the directory that you want to share with the container. In "Container path," enter `/workspace`

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1. **Launch the container** from Docker Desktop:
   - Go to the "Images" tab.
   - Click the play icon next to the `class-docker` image.
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     - **Ports**: Map "Container port" 22 to "Host port" 2222.
     - **Volumes**: In "Host path," enter the directory that you want to share with the container. In "Container path," enter `/workspace`

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1. **Launch the container** from Docker Desktop:
   - Go to the "Images" tab.
   - Click the play icon next to the `class-docker` image.
   - Set the following options:
     - **Name**: `class-container`
     - **Ports**: Map "Container port" 22 to "Host port" 2222.
     - **Volumes**: In "Host path," enter the directory that you want to share with the container. In "Container path," enter `/workspace`

Setting a Volume is important because only files stored in that directory will survive if the container is destroyed!

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```
docker run -dit \
    --name class-container \
    --publish 2222:22 \
    --volume /home/user/cogs205b:/workspace \
    class-docker
```

Step 2: Run the Docker Container

2. Alternatively, you can **use the terminal** to run the container:

```
docker run -dit \
    --name class-container \
    --publish 2222:22 \
    --volume /home/user/cogs205b:/workspace \
    class-docker
```

(But don't do this now, use Docker Desktop.)

Step 2: Run the Docker Container

3. **Verify the container is running**:

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   - In Docker Desktop, check that the `class-container` status shows as "Running."

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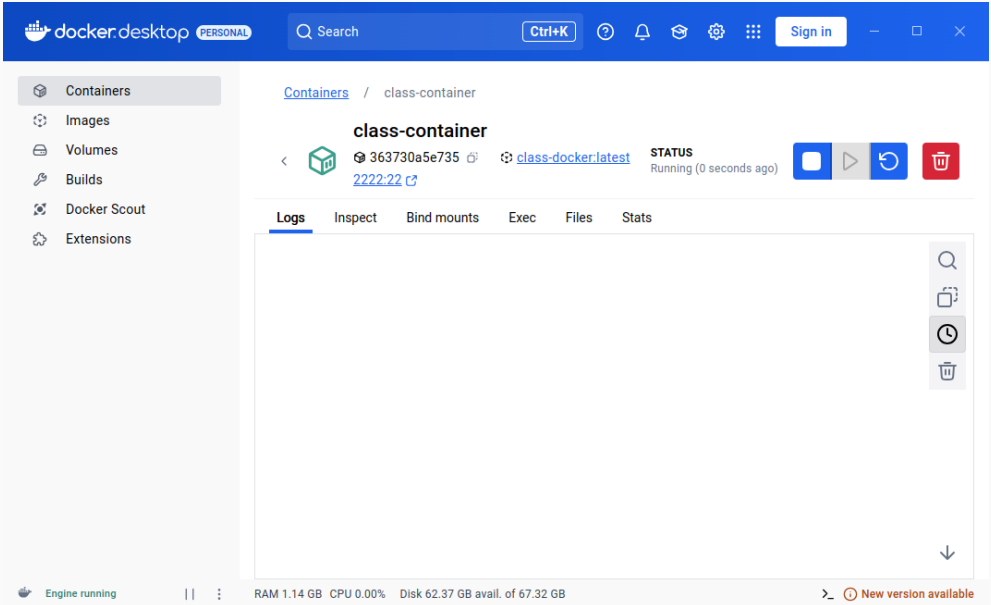
```
docker ps
```


docker desktop PERSONAL
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Ctrl+K
Sign in
Containers
Images
Volumes
Builds
Docker Scout
Extensions
Containers / class-container
class-container
363730a5e735
class-docker:latest
2222:22
STATUS
Running (0 seconds ago)
Logs
Inspect
Bind mounts
Exec
Files
Stats
Engine running
RAM 1.14 GB CPU 0.00% Disk 62.37 GB avail. of 67.32 GB
New version available

Step 3: Test the container

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1. Use Docker Desktop's integrated shell, and test the Python installation:

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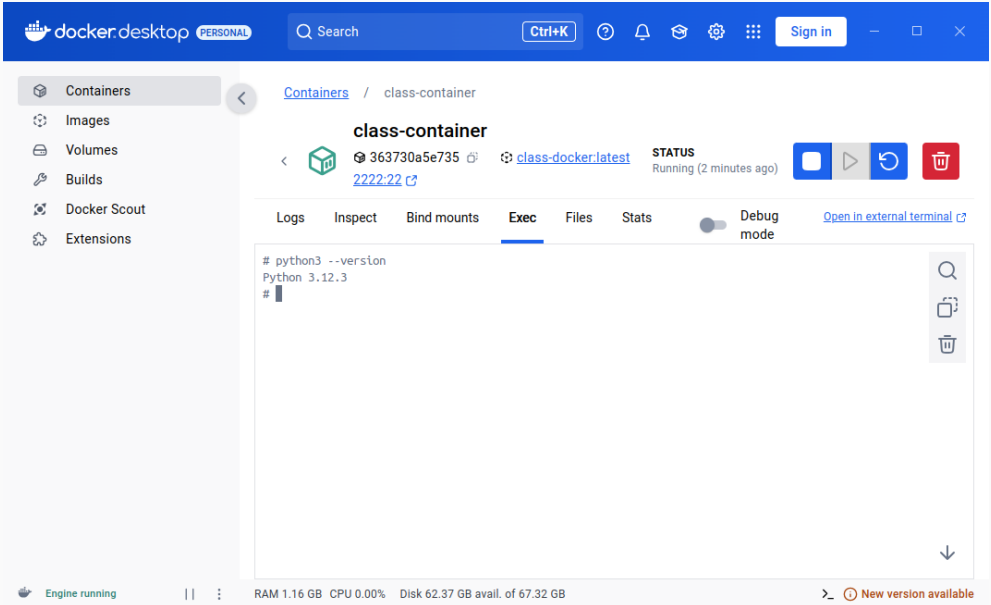
1. Use Docker Desktop's integrated shell, and test the Python installation:

```
python3 --version
```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Containers / class-container
class-container
363730a5e735
class-docker:latest
2222:22
STATUS
Running (2 minutes ago)
Logs
Inspect
Bind mounts
Exec
Files
Stats
Debug mode
Open in external terminal
python3 --version
Python 3.12.3
#
Engine running
RAM 1.16 GB CPU 0.00% Disk 62.37 GB avail. of 67.32 GB
New version available

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The integrated shell is running on the Ubuntu/Linux operating system inside the container. You can learn the basic commands of the command line interface from the cheat sheet I'll put on GitHub.

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The integrated shell is running on the Ubuntu/Linux operating system inside the container. You can learn the basic commands of the command line interface from the cheat sheet I'll put on GitHub.

Alternatively, open your system's terminal to connect to the container:

```
docker exec -it class-container bash
```

When you're done: Stop (and later restart) the container

In Docker Desktop, click the stop or play button in the Containers tab.

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Or, in the terminal to stop the container:

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Or, in the terminal to stop the container:

```
docker stop class-container
```

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docker stop class-container
```

In the terminal to restart the container:

When you're done: Stop (and later restart) the container

In Docker Desktop, click the stop or play button in the Containers tab.

Or, in the terminal to stop the container:

```
docker stop class-container
```

In the terminal to restart the container:

```
docker start class-container
```

Summary

You've now set up the class Docker container, confirmed it works, and learned how to manage it using Docker Desktop. This container will be your consistent Python environment for all class activities.

Summary

You've now set up the class Docker container, confirmed it works, and learned how to manage it using Docker Desktop. This container will be your consistent Python environment for all class activities.

In the future, you could make new containers for new projects and manage them in Docker Desktop or submit them to Docker Hub.